Privacy Shield:

Secure & Private Fraud Detection Against Model Inversion Attacks

Course: Security and Privacy for Machine Learning
Course Code: 216H17C701

1. Problem Statement

Machine learning is widely used to identify fraudulent financial transactions. However, financial transaction data contains sensitive information, and the trained ML model may also leak information about that data. An attacker with access to the model may attempt to infer sensitive information from its outputs. The proposed project therefore aims to build a fraud detection model that applies multiple privacy-preserving techniques while maintaining useful fraud-detection performance.

2. Objectives

- Develop an ML model for classifying financial transactions as legitimate or fraudulent.
- Study privacy leakage using one attack: Model Inversion Attack (MIA).
- Apply Local Differential Privacy to selected sensitive input attributes.
- Apply Differential Privacy during model training.
- Protect the input data using privacy-preserving data modification.
- Apply Compressive Privacy to the feature representation.
- Evaluate the trade-off between privacy protection and fraud-detection utility.

3. Use-Case Justification

The selected use case is financial transaction fraud detection in digital banking or financial services. It is directly related to cybersecurity because fraud is a security threat and ML is used to detect suspicious transactions. Transaction information can also contain sensitive attributes, making the use case suitable for studying privacy-preserving machine learning. The use case is intentionally different from healthcare, retail, and smart-home applications.

4. What the Model Is About

Privacy Shield is a privacy-preserving binary classification system. It receives transaction features such as amount, transaction category, time-related information, channel/device information, and other suitable features available in the selected dataset. It predicts whether a transaction is legitimate or fraudulent.

Transaction → Privacy Protection → Feature Representation → MLP → Fraud / Legitimate

5. Attack: Model Inversion Attack

The project uses only one attack: Model Inversion Attack. The attacker has query access to the trained fraud-detection model and observes its prediction outputs. The attacker attempts to infer sensitive information about a transaction or construct a plausible input representation that produces a desired model response.

The attack is applied to the baseline model and the privacy-preserving model so that the effect of the privacy techniques can be measured.

6. Threat Model

The protected asset is the sensitive information contained in or represented by the training data and learned model. The attacker does not directly receive the private training records but can query the target model and observe its outputs. The attacker attempts to recover or infer sensitive input information. Privacy Shield aims to reduce this leakage without making the fraud classifier unusable.

7. Workflow of the Proposed System

The complete workflow is:

Step	Component	What happens
Step 1	Financial transaction data	Collect/select a suitable financial fraud dataset.
Step 2	Fraud prediction	The trained model predicts whether a transaction is legitimate or fraudulent.
Step 3	Module 1: Threat/Attack evaluation	Perform the Model Inversion Attack against the trained model to test whether sensitive information can be inferred.
Step 4	Module 4: Privacy-preserving data mining	Remove direct identifiers and modify/generalize sensitive or quasi-identifying input attributes.
Step 5	Module 3: Local Differential Privacy	Randomize selected sensitive attributes locally before they are used by the central ML pipeline.
Step 6	Module 5: Compressive Privacy	Transform/compress the protected feature representation to reduce unnecessary information exposure while retaining useful predictive information.
Step 7	Module 2: Differential Privacy	Train the supervised MLP fraud classifier using a differentially private training mechanism such as DP-SGD.
Step 8	Fraud prediction	The trained model predicts whether a transaction is legitimate or fraudulent.
Step 9	Module 1: Threat/Attack evaluation	Perform the Model Inversion Attack against the trained model to test whether sensitive information can be inferred.

Step 10	Privacy–utility evaluation	Compare fraud-detection performance with the attack/inference performance and privacy parameters.

The five modules are not treated as five separate mini-projects. They form one continuous privacy-and-security workflow. Each module addresses a different stage or aspect of the same fraud-detection system.

Module 1 → Module 4 → Module 3 → Module 5 → Module 2 → Model → Module 1

Module 1: Threats and Attacks for ML Systems
A Model Inversion Attack is used to test whether the model reveals sensitive information on a baseline MLP

Module 4: Privacy-Preserving Data Mining
This is the first protection layer. The input data is cleaned and modified before model development. Direct identifiers are removed, and sensitive or quasi-identifying attributes can be generalized or otherwise modified. This reduces unnecessary exposure before ML processing.

Module 3: Local Differential Privacy
After the initial privacy-preserving data preparation, selected sensitive attributes are randomized using an LDP mechanism. The important idea is that the sensitive value is protected before entering the central ML processing pipeline.

Module 5: Compressive Privacy
The LDP-protected data is transformed into a compressed feature representation. The purpose is to reduce the amount of detailed information carried by the representation while retaining enough information for fraud classification. The effect on utility is measured experimentally.

Module 2: Differential Privacy
The compressed, privacy-protected representation is then used to train the supervised fraud classifier with Differential Privacy. A mechanism such as DP-SGD limits the influence of individual training records on the learned model.

Module 1: Threats and Attacks for ML Systems
Finally, the resulting model is treated as a security target. A Model Inversion Attack is used to test whether the model still reveals sensitive information. This closes the workflow by measuring the actual privacy risk after applying the protection layers.

Thus, the workflow follows a logical progression: protect the data → protect local values → reduce feature exposure → protect model training → attack the model to evaluate the remaining privacy risk.

8. Module-to-Project Mapping

Course Module	Selected Topic	Role in Privacy Shield
Module 1	Threats and Attacks for ML Systems	Provides the security threat and Model Inversion Attack used to evaluate the model.
Module 2	Differentially Private Supervised Learning	Protects the model-training process using Differential Privacy.
Module 3	Local Differential Privacy and LDP mechanisms	Protects selected sensitive attributes before central ML processing.
Module 4	Protecting privacy by modifying the input	Reduces sensitive information in the input before ML processing.
Module 5	Compressive Privacy and accuracy of the utility task	Compresses/transforms features and measures the resulting privacy–utility trade-off.

This mapping follows the five modules listed in the uploaded Security and Privacy for Machine Learning syllabus, including the topics on ML attacks, differential privacy, LDP, privacy-preserving data mining, and compressive privacy.

9. Evaluation

The system will be evaluated from two perspectives:

- Utility: Accuracy, Precision, Recall and F1-score for fraud detection.
- Privacy/Security: Model Inversion Attack success or inference quality, together with the relevant privacy parameters.
- Comparison: Baseline MLP versus the privacy-preserving pipeline.
- Trade-off: Determine how increasing privacy protection affects fraud-detection performance.

10. Expected Outcome

The project is expected to produce an end-to-end privacy-preserving fraud-detection workflow in which all five course modules contribute to the same system. The final experiments will determine whether the combined privacy mechanisms can reduce model inversion risk while retaining acceptable fraud-detection performance.